



**Optimizing virus filtration for continuous processing using
serial filtration at high area ratio**

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Optimizing virus filtration for continuous processing using serial filtration at high area ratio

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ABSTRACT

Compared to batch operation, continuous bioprocessing can offer numerous advantages, including increased productivity, improved process control, reduced footprint, and increased flexibility. However, integration of traditional batch operations into a connected process can be challenging. In contrast to batch operations run at constant pressure or high flux, virus filtration in continuous processes may be operated at very low flux. This change in operating conditions may reduce the viral retention performance of the filter which has inhibited adoption of truly continuous virus filtration. To overcome this limitation, a novel approach is described that utilizes serial virus filtration, with a high area ratio between first to second stage filters, to achieve virus retention targets. In this study, virus filters were operated continuously (except for planned process interruptions) for 200 hours in a serial configuration at a first to second stage filter area ratio of 13:1 and at a first stage flux of 5 L/m²/hr. While the minute virus of mice (MVM) retention performance of the first stage filter was about 4 log reduction value (LRV), there was no virus detected in the second stage filtrate, translating to an MVM LRV across the filtration train of ≥6.7. The second stage filter was the dominant flow resistance at the start of the run but, as it was protected from foulants by the first stage filter, it suffered minimal fouling and the life of the filter train was controlled by the first stage. A theoretical case study projected that continuous virus filtration using serial configuration at high area ratio would have about 30% longer filter changeout time, 14% higher productivity, and virus retention nearly 6 LRV greater than single stage operation. The findings of this research are expected to provide valuable insights into optimizing virus filtration in continuous bioprocessing.

KEYWORDS: Virus Filter, Virus Filtration, Virus Clearance, Continuous Processing

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1. Introduction

Continuous bioprocessing has gained significant attention in the biopharmaceutical industry due to its potential benefits such as increased productivity, enhanced process control, reduced footprint, and improved flexibility compared to traditional batch operations (Konstantinov et al., 2015; Drobnjakovic et al., 2023; Gerstweiler et al., 2021). However, the successful integration of batch operations into a continuous process poses certain challenges. One particular area of concern is virus filtration, a critical step in bioprocessing to ensure the removal of viral contaminants from the final product. In batch processes, the virus filter is commonly operated at high flux so that the filtration of the batch is completed within a few hours. In contrast, the duration of the virus filtration process in a continuous process could be several days or more. Consequently, to filter the same quantity of fluid but spread out over a longer time frame, the flux through a virus filter in a continuous process will be much lower – by a factor of 10 or more – compared to batch processes (Bohonak et al., 2021; Coffman et al., 2021; Kozaili et al., 2023; Gerstweiler et al., 2021; Shirataki et al., 2023; Coffman et al., 2021b; David et al., 2023).

The impact of low flux operation on virus retention performance depends on numerous factors, with some reports showing minimal impact (Bohonak et al., 2021; Kozaili et al., 2023; McAlister et al., 2019) and others a more significant effect (David et al., 2019; Fan et al., 2021; Yamamoto et al., 2014; Gustafsson et al., 2018; Strauss et al., 2017; Afzal & Zydney 2022). A commonly proposed mechanism is that diffusion of virus particles at lower flux increases the probability that virus particles are able to migrate to and pass through larger pores (Fan et al., 2015; Yamamoto et al., 2014; Leisi et al., 2021). The concern that lower flux rates may hinder the ability of the filter to effectively capture and retain viral particles has complicated the implementation of continuous virus filtration. As a result, while there are reports of successful demonstrations of continuous virus filtration (Bohonak et al., 2015; Shirataki et al., 2023; Lute et al., 2020; Arnold et al., 2019; Kozaili et al., 2023), there has been limited adoption of truly continuous virus filtration in industrial-scale bioprocessing (Konstantinov et al., 2015).

Operation of virus filters at low flux within a continuous process can be avoided by operating the filters for only a few hours a day at high flux, essentially as a daily batch process within a continuous process (Coffman et al., 2021; Coffman et al., 2021b; Leisi et al., 2021). A single filter position can be used, or several filters can be arranged in parallel and the system switches from one filter to another after the filter life has been exhausted, but either way periodic filter changeouts are required within the continuous process (Gerstweiler et al., 2021). Also, to accommodate the sub-batch virus filtration operation, surge tanks are used upstream and downstream of the virus filter, which adds cost and complexity to the process. The required daily changeout of the virus filter and the requirement for surge tanks reduce the advantage of continuous processing. A solution that allows for truly continuous virus filtration without compromising virus retentive performance is therefore preferred.

A method to boost virus clearance and minimize the risk of a virus retention shortfall is to operate two virus filters in series (Kozaili et al., 2023; Nowak et al., 2017). In such an arrangement, viruses that escape capture in the first stage virus filter will likely be retained in the second stage filter. The retentiveness of a membrane is often expressed as either a sieving value, C_p/C_f , or as a log reduction value:

$$LRV = \log_{10} \left[\frac{C_f}{C_p} \right] \tag{1}$$

where C_f is the concentration of the retained species in the feed and C_p is the concentration of the retained species in the permeate (a.k.a. filtrate). The total LRV of filters operated in series is the sum of the LRV of each filter (McAlister et al., 2019; Leisi et al., 2021). If low flux operation imposes a risk of decreased LRV, that risk can be mitigated by adding a second stage filter. A drawback of this approach, however, is the added cost of a second stage filter. Since virus filtration can be one of the more expensive downstream processing steps, this downside is not an insignificant consideration (De Souza et al., 2016). In addition, the second stage filter will impose added flow resistance that at constant flux operation will translate to a higher pressure drop across the filtration train. Since the inlet pressure to the filter(s) may be constrained by equipment limitations, this added pressure loss could limit full utilization of the filters. If the virus filters experience fouling, the pressure drop across the filters will increase with increasing throughput and the filtration endpoint will be the point at which the maximum allowable inlet pressure is reached. The added flow resistance of an added stage of filtration will decrease the amount of pressure rise available to the filters before they reach the system limit.

As discussed above, adding a second filtration stage will improve the overall virus retention but at the cost of additional filters. At a first to second stage filter area ratio of 1:1, the cost of the filters will be doubled compared to single stage operation. However, this added cost can be reduced by operating at higher area ratios. At an area ratio of 10:1, for example, the added filter cost compared to single stage operation may be as little as 10%. Another advantage of a high area ratio configuration is that the second stage will operate at a proportionately higher flux than the first stage. Using the 10:1 area ratio example, if the first stage is operated at a flux of 5 L/m²/hr (LMH), then the flux through second stage filter will be 50 LMH. This means that the second stage filter will be much less susceptible to diminished virus retention performance associated with low flux operation.

The added flow resistance of the second stage will be exacerbated at high area ratio, and especially for virus filters that have low permeability this added flow resistance may be prohibitive. But for virus filters that have a sufficiently high permeability, and in situations where the first stage filtration is at very low flux, it may be feasible to accommodate even very high area ratio configurations. Since the second stage filter will be largely protected from foulants by the first stage filter, it should experience minimal plugging. Therefore, while the second stage filter will initially be responsible for the bulk of the filter train flow resistance, as the first stage filter plugs the second stage filter will become a decreasing fraction of the total train resistance and the filtration endpoint will be controlled by the flow resistance through the first stage filter.

In this study, the performance of continuous virus filtration using a serial filtration arrangement at high first to second stage filter area ratio was investigated and compared to single stage operation. The objective of the study was to assess filtration performance and virus clearance across a high area ratio serial filtration train at conditions that are representative of continuous processes, particularly at a low flux condition where there may be a risk of diminished virus retention performance. The effect of process pause during virus filtration, a condition that has been reported to result in elevated virus breakthrough (Leisi et al., 2021; LaCasse et al., 2013; Dishari et al., 2015), was also investigated within a continuous virus filtration process.

The findings of this research are expected to provide valuable insights into optimizing virus filtration in continuous bioprocessing. The demonstrated robustness, improved virus retention performance, and increased filter productivity associated with the serial virus filtration strategy have the potential to enable wider adoption of continuous virus filtration in industrial-scale biopharmaceutical manufacturing. By enhancing the efficiency and reliability of virus filtration, this work contributes to ensuring the safety and quality of biopharmaceutical products, ultimately benefiting patients and the bioprocessing community.

2. Theoretical Background

2.1 Simulation of Serial Virus Filtration Performance

There are several performance parameters that contribute to the overall performance of a virus filter. These include permeability, throughput capacity and virus retention. All of these performance parameters can depend on the operating conditions and filter configuration. In the following sections, how each of these performance parameters within each stage of a serial filtration arrangement impact the performance of a filtration train will be briefly elucidated.

2.1.1 Permeability

The flow resistance of filters operated in series is the sum of the resistances of each of the filters in the filtration train. For a two-stage filtration train, the total resistance in terms of trans membrane pressure per volumetric flow rate - neglecting any flow resistances associated with inlet and outlet ports and any external plumbing - is simply:

$$R_t = R_1 + R_2 \quad (2)$$

Where

R_t = total resistance

R_1 = stage 1 resistance

R_2 = stage 2 resistance

Within a stage, the resistance is inversely proportional to effective filtration area (EFA). If the unfouled properties of the membrane in each stage are the same, at the start of the filtration Eq. 2 can be rewritten as:

$$R_t = (\frac{A_1}{A_2} + 1)R_1 \tag{3}$$

Where

A_1 = filtration area in stage 1

A_2 = filtration area in stage 2

The dimensionless quantity A_1/A_2 is the first to second stage area ratio. Permeability in terms of volumetric flow rate per unit filter area per unit pressure is the inverse of $A \cdot R$. For a given flow resistance of the first stage and a given first to second stage area ratio, the total flow resistance of the filtration train (neglecting any flow resistances associated with inlet and outlet ports and any external plumbing) is easily determined.

2.1.2 Throughput Capacity

Throughput capacity of a filter is defined as the volumetric throughput per effective filtration area of a filter before it reaches the end of its useful life. The filtration endpoint can be based on either a maximum operating time, a maximum mass or volumetric throughput, a minimum flow rate through the filter at a specified inlet pressure, a maximum pressure drop across the filter at a specified flow rate, or a maximum level of fouling or filter permeability decay relative to a clean filter. For virus filtration, a maximum level of flux (or permeability) decay is commonly used to establish the filtration endpoint (Bolton et al., 2005). There are numerous plugging mechanisms and associated mathematical models of varying levels of sophistication and complexity that describe fouling behaviors in filters (Iritani, 2013). For simulating flow resistance as a function of volumetric throughput of a filter within a filtration train, a simple single mechanism blocking model was selected from among the four classical complete blocking, standard blocking, intermediate blocking, and caking models (Hermia, 1985). In many cases, the intermediate model, in which the foulants reaching the membrane directly block some membrane area or deposit on previously deposited particles, can provide a good fit to virus filtration data, at least before the onset of caking at the point where the membrane is heavily fouled (Peles et al., 2023). For the intermediate model, flow resistance R as a function of throughput volume can be expressed as:

$$\frac{R}{R_0} = e^{k_i V} \tag{4}$$

Where

R_0 = unfouled filter resistance

k_i = intermediate blocking constant

V = volume filtered per filtration area

The pressure drop across the filter can be calculated using:

$$\Delta P = R * J * A \tag{5}$$

Where

ΔP = Pressure drop across the filter

J = Flux

A = filter area

The intermediate model can be applied to both filtration stages but as will be shown later, because the first stage filter captures nearly all the particles that could potentially foul the second stage filter, fouling of the second stage filter can be considered practically negligible. Filter resistance as a function of volume filtered within each stage can be estimated using Eq. 4, and the total resistance across the filtration train can be calculated using Eq. 3.

2.1.3 Virus Retention

Virus retentive filters function primarily by size exclusion (Virus Retentive Filtration, 2022). That is, viruses are unable to traverse membrane passageways in which the apertures are smaller than the virus. However, within the upper end of the pore size distribution of a virus filter, there can be a small number of passageways that are large enough to allow some viruses to escape through. While the probability of a virus encountering such passageways is very low (by definition, for a virus filter with an LRV of 4, only 0.01% or fewer of the viruses entering the filter will pass through the filter), the probability of passage can depend on the operating conditions. As noted earlier, at low flux conditions diffusive transport can become more prominent than convective transport. In contrast to convective transport in which the movement of a virus is controlled by the flow streams of the liquid and can be forced to be captured at a filter chokepoint, diffusive transport can allow for the opportunity for the virus to meander within the membrane structure and eventually encounter a passageway through which the virus can exit the filter into the filtrate. The impact of flux on virus LRV depends on the membrane structure, fluid properties, virus size, and other factors and while in theory can be modeled based on first principles, in this paper the effect of flux on LRV for serial filtration modeling purposes relies on only empirical data and will be further elucidated in the methods and results sections.

Another factor that can impact virus retention is membrane fouling. The effect of fouling on LRV depends on the fouling mechanism and filter pore structure, but if foulants preferentially block the smaller pores within the filter pore size distribution then a greater proportion of the flow will be directed to the larger pores, which have a higher probability of allowing virus passage, resulting in a lower LRV. According to the model developed by Bolton (Bolton et al., 2005), LRV as a function of flux decay can be described by

$$LRV = LRV_0 - \text{Log}\left(\frac{R}{R_0}\right) \quad (6)$$

Where LRV_0 is the LRV of the unfouled membrane. Substituting Eq. 4 into Eq. 6, LRV as a function of volumetric throughput can be expressed as:

$$LRV = LRV_0 - \text{Log}(e^{k_i V}) \quad (7)$$

As noted earlier, the total LRV of filters operated in series is the sum of the LRV of each filter. The LRV of the filtration train is therefore:

$$LRV_t = LRV_{0,1} + LRV_{0,2} - (Log(e^{k_{i,1}}V_1) + Log(e^{k_{i,2}}V_2)) \tag{8}$$

To reiterate, the sensitivity of retention to degree of membrane fouling is dependent on fouling mechanism, membrane morphology (pore structure), and operating conditions. The impact of fouling on retention can vary widely among different virus filters even at the same operating conditions and degree of fouling so the equations in this section relating LRV to fouling should be considered to represent just one potential conservative scenario among others.

3. Materials and Methods

3.1 Virus Filters

Three commercially available Viresolve® Pro Device virus filters supplied by MilliporeSigma (Burlington MA) were used in this study, listed in Table 1. These filters contain two layers of asymmetric polyethersulfone (PES) membrane and are operated in normal flow filtration mode.

Table 1. Viresolve® Pro Device: filter formats and filtration areas.

Filter Name	Effective Filtration Area (m²)
Micro 40	0.00034
Modus 1.1	0.017
Modus 1.3	0.22

3.2 Feed Stream Preparation

Feed material was prepared by adding human immunoglobulin (hIgG) (SeraCare Life Sciences, HS-470-90) in 50 mM sodium acetate, 100 mM sodium chloride, pH 5 to 6 g/L and then prefiltering using an Opticap XL10 Viresolve® NFR capsule (MilliporeSigma, KZRVA10TT1). The 6 g/L hIgG material was further diluted in sodium acetate buffer to a target concentration depending on the experiment. Challenge feed materials and buffers were sterile filtered using Stericup® filters (MilliporeSigma, S2GPU10RE) or Steripak™-GP filters (MilliporeSigma, SPGPM10RJ) and were held at room temperature for the duration of each study. For evaluating the impact of flux on virus retention, challenge feed was prepared by spiking MVM to 6.3 log TCID₅₀/mL into 0.07 g/L IgG. For the continuous serial filtration experiments, virus challenge feeds consisting of 0.05 g/L IgG spiked with MVM to 7 log TCID₅₀/mL were prepared fresh at 0, 96, and 167 hours of filtration time. This IgG concentration was selected to be representative of a fouling stream for which a throughput of 500-1000 L/m² would result in about 90% flux decay (V90) relative to the clean membrane.

3.3. Single Stage LRV vs Flux

Viresolve® Pro Micro 40 filters were wetted with Milli-Q® water at the manufacturer's recommended conditions. The wetted devices were connected to 20-, 30-, and 50-mL syringes, (Becton Dickinson, 302830, 302832, 309653) depending on the target flux, filled with MVM-spiked IgG feed and loaded onto a syringe pump (New Era, model 1600). The filters were first operated at an elevated flux for at least 25 L/m² before adjusting the flux down to the desired set point. At each of the 10 flux set points between 5 and 117 LMH, at least 25 L/m² of fluid of filtrate was processed before collecting filtrate samples for virus assays.

3.4 Serial Virus Filtration

Viresolve® Pro Modus 1.3 and Viresolve® Pro Modus 1.1 filters were employed as the stage 1 and stage 2 filters in the serial filtration set-up (Figure 1). The inlet before the peristaltic pump (Masterflex® L/S®, Model 07522-20) comprised two lines, non-spiked IgG feed and virus-spiked IgG feed. A three-way valve was used to switch between the non-spiked and spiked feed lines. Samples of the virus challenge feed were collected at 24-hr intervals and assayed to measure virus viability loss over the duration of the test and determine the virus challenge load over time. Tee connections were used at the first and second stage filtrates to collect samples continuously. The two multichannel syringe pumps (Cole Parmer®, Model 75900-50 and KD scientific, Model 210) were used to draw samples for virus assays. 12 mL samples were collected over 24 hr and additional samples were collected immediately after process interruptions and when virus challenge feed was introduced. Samples were collected at a flow rate such that <1% of the filtrate flow was pulled into the syringe.

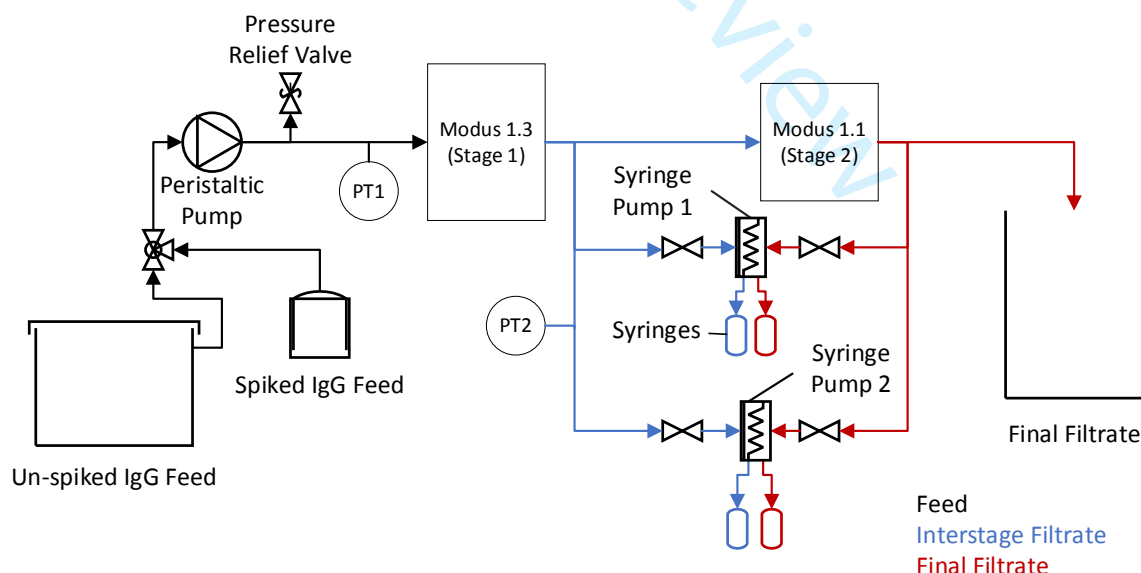


Figure 1. Experimental setup for continuous serial filtration. Syringe pump 1 was used to collect 24-hr samples and syringe pump 2 was used to collect process interrupt and post-spike samples. Pressure was measured at the inlet with pressure transducer 1 and interstage with pressure transducer 2.

PendoTech-Single use pressure sensors (PREPS-N-000) were placed inline to measure the inlet and interstage pressures and assess the rate of filter fouling. A 138 kPa total pressure end point was selected as the system limit. A pressure relief valve at the inlet of the manifold prevented the system from exceeding 145 kPa.

The Modus® 1.1 and 1.3 filters were initially wetted with Milli-Q® water at the manufacturer’s recommended conditions. MVM spiked feed was introduced to the stage 1 device, vented to remove air bubbles. The peristaltic pump was set to 5 LMH for the first stage filter, and the continuous filtration was initialized. After challenging with the virus feed for 4 hr, the inlet was switched to the non-spiked 0.05 g/L IgG feed. Virus challenge feed was introduced at 0, 97, 169 hr with a duration of 4 hr before switching to the non-spiked IgG feed line. Process interrupts, in which the serial filtration system was stopped for 1 hr, were introduced after 96, 145 and 168 hr. The 24-hr, post-virus challenge, and post-process interrupt filtrate pools were frozen at -80°C until assayed.

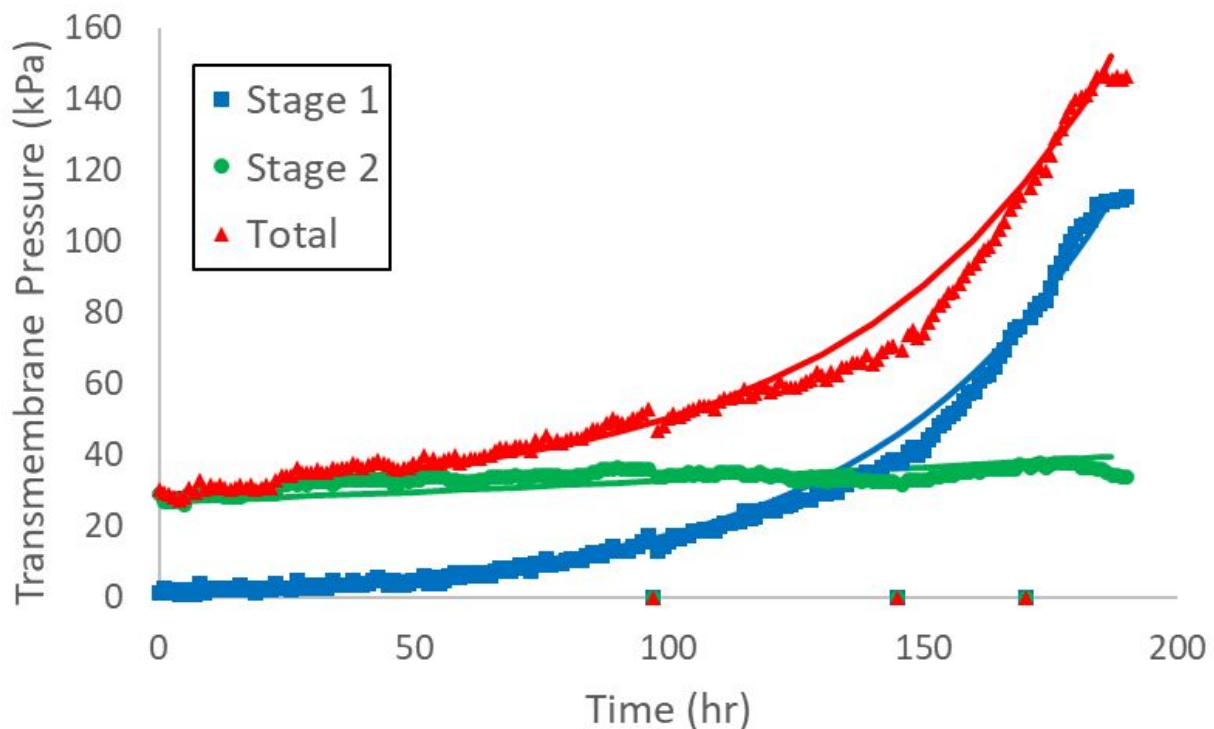
3.3 Virus Quantification

Mammalian virus MVM (ATCC® VR-1346™), was propagated using internal protocols to generate purified high titer stocks. MVM titer was determined using a 50% Tissue Culture Infectious Dose assay using the indicator cell line 324k (P. Tattersall, Yale University). Samples were serially diluted in growth media, Dulbecco’s Modified Eagle Medium (ThermoFisher, 11965092) supplemented with 1% of the following: fetal bovine serum (ThermoFisher, 100099141), 200 mM L-glutamine (ThermoFisher, 25030081), 100× MEM nonessential amino acids (ThermoFisher, 1140035), and 10,000 units/mL penicillin–streptomycin (ThermoFisher, 15070063). Serial dilutions were plated into a 96-well plate seeded with the indicator cell line and incubated at 37 °C for 10 days. After incubation, the plates were visually scored to identify infection, and the resulting ratios of infected wells to total wells for each dilution were used to calculate virus concentration using the Spearman-Kärber method (Kärber, 1931). The cumulative LRV at any time point was calculated by subtracting the log₁₀ of the total virus measured in the filtrate fractions from the log₁₀ of the total virus entering the filters.

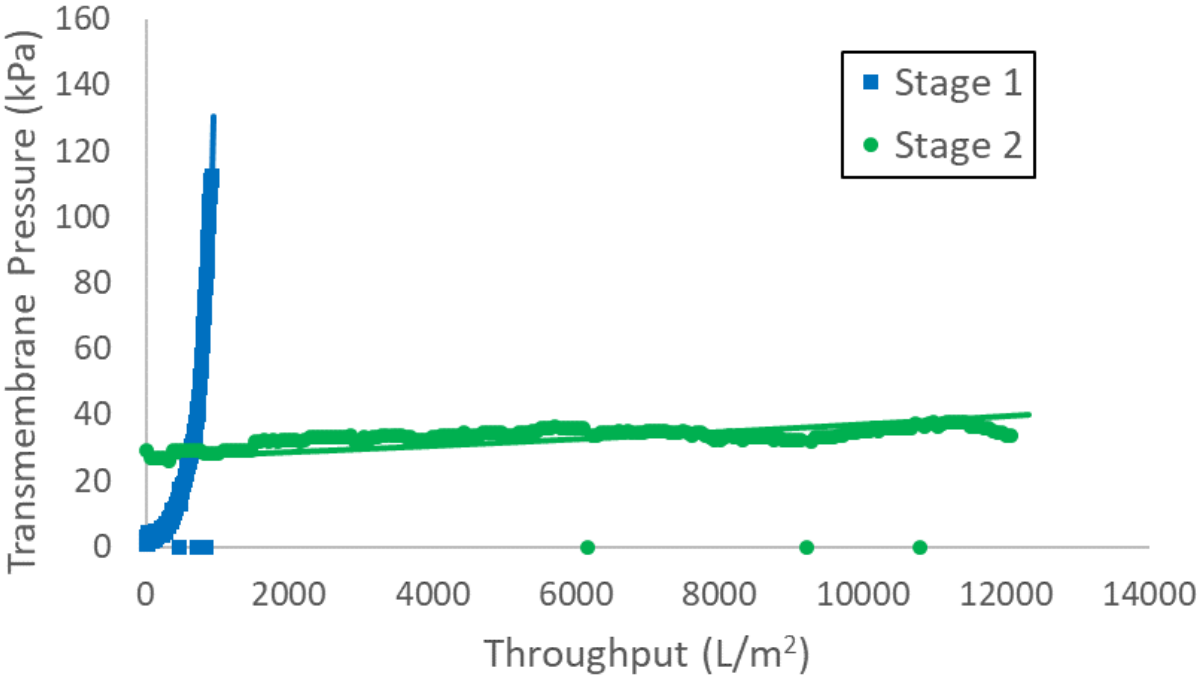
4. Results and Discussion

4.1 Throughput capacity

Pressure differential across each filter as a function of filtration time is shown in Fig. 2a and as a function of volumetric throughput in Fig. 2b. Initially, the pressure drop across the filtration train (designated as “total” in Fig. 2b) was dominated by the pressure drop across the second stage filter. This was expected (Eq. 3) since it was operating at a flux of 65 LMH which was 13 times higher than that of the first stage. As the filtration progressed, the first stage membrane fouled and the transmembrane pressure (TMP) increased. The second stage filter, which was protected from foulants by the first stage filter, experienced minimal fouling and minimal TMP increase. A fit of the data to the intermediate model found that the blocking constant for stage 1 was more than two orders of magnitude higher than for stage 2. Flow resistance for stage 1 increased by a factor of 10 (V90) compared to the initial resistance at about 530 L/m². This compares to a model fit projected V90 value of about 70,000 L/m² for the stage 2 filter.



(a)



(b)

Figure 2. Transmembrane pressure as a function of (a) filtration time; (b) volumetric throughput. Solid lines are fits of the data to the intermediate model with R_0/A for both stages = 0.42 kPa/LMH, $K_{i,1} = 4.35$, and $K_{i,2} = 0.033$. For about the last 5 hours of data, the total TMP was limited to 138 kPa by a pressure relief valve forcing a lower flux so this data was not included in the model fit. Data points at zero TMP were at process interrupts when the flow to the filters was stopped and likewise were excluded from the model fit.

Although the second stage filter was initially responsible for the majority of the filtration train pressure drop, the filtration endpoint was controlled by the pressure drop across the first stage filter. The filtration train pressure drop reached 138 kPa at a throughput volume of about 890 L/m² of first stage area, or about 825 L/m² of total filtration area. At that point, about 70% of the total filtration train pressure drop was through the first stage, compared to about 7% at the start of the run. Using the fit of the intermediate model to the first stage data, it is projected that the ΔP would reach 138 kPa at a throughput of about 965 L/m² (and also reaching ~98% permeability decay). Thus, from a total pressure drop perspective, the second stage limited the capacity of the first stage by only about 8%.

However, this capacity limitation can be more than offset by a couple of factors. First, as noted, the filtration endpoint can be constrained by an upper limit on permeability decay. If a maximum permeability decay of 90% is imposed in this case (to minimize the risk of virus retention performance degradation), then the filtration train ΔP would only be about 62 kPa (of which about 28 kPa is contributed by the first stage) at that point so the second stage ΔP would have no effect on the useful life of the first stage filter. In fact, since the second stage adds a significant layer of retention assurance, it can allow an extension of the first stage life since the overall risk of a virus retention shortfall is significantly minimized. In this study, the first stage permeability decay reached about 98% where the filtration train ΔP reached the 138 kPa upper limit. As noted above, the combined throughput capacity was about 825 L/m² at that point, which is about 55% higher than the V90 value of the first stage only. If the first stage was limited to V95, the total filter productivity with serial operation would be ~20% higher compared to single stage operation with a V90 filtration endpoint constraint. Additionally, since the second stage filter experienced only minimal permeability decay, it could remain in use after changeout of the first stage filter. This point will be discussed further in the process simulation section.

4.2 Virus retention

The effect of flux on Viresolve® Pro Device MVM clearance performance as measured over a range from 5 to 117 LMH is shown in Figure 3. The line between 5 and 60 LMH is a linear fit to that subset of the data. Above 60 LMH, a horizontal line is shown that represents the assay limit of the retention test. As seen in Figure 3, below about 60 LMH LRV decreases with decreasing flux whereas above 60 LMH there is not a measurable impact of flux on LRV. In this serial filtration study, the first stage was operated at 5 LMH, where the impact of low flux on LRV is significant, and the second stage was operated at 65 LMH, where there is low sensitivity to flux on LRV.

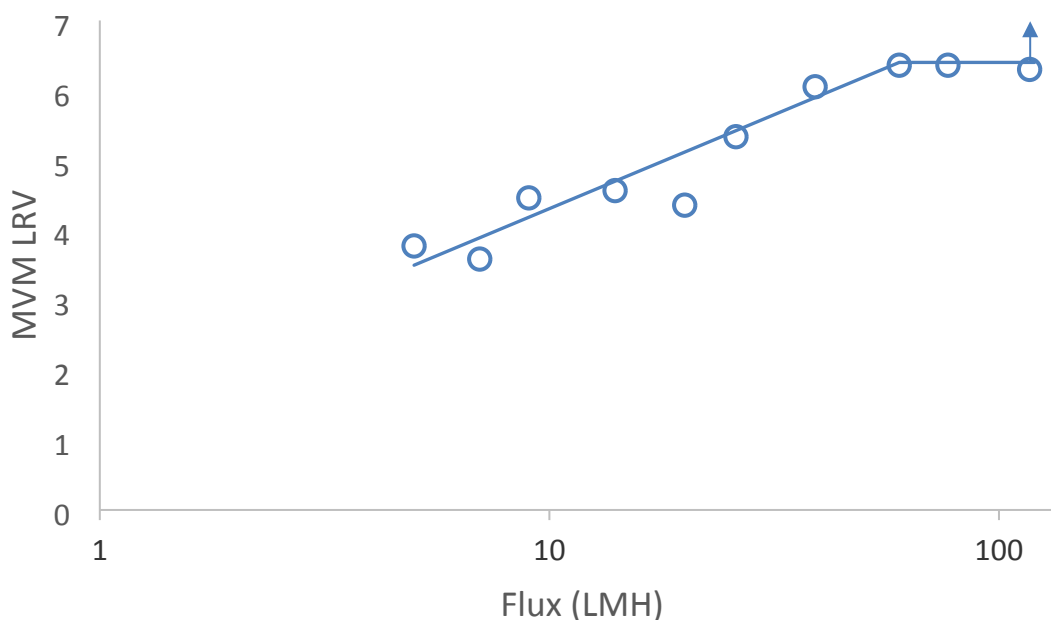


Figure 3. Effect of flux on MVM clearance performance (LRV) on Viresolve® Pro Micro 40 devices. Arrow indicates that no virus was detected in the filtrate and that the LRV is a minimum.

For the serial filtration study, cumulative total MVM in the first stage inlet, interstage, and second stage filtrate streams as a function of operating time is shown in Figure 4 and LRV across the first stage and across the filtration train is shown in Figure 5. The MVM challenge load retained infectivity (within the 95% confidence interval of the virus assay of $\sim\pm 0.3$ log TCID₅₀/mL) over the duration of the study and the challenge load increased with subsequent MVM spikes. The MVM clearance performance of the first stage virus filter, which at any point in time is the total MVM (log TCID₅₀) that challenged the filter minus the total MVM in the stage 1 filtrate, was in the range of 3.8–4.6 LRV. The stage 1 filter virus clearance performance was slightly higher than expected based on low flux range study (Figure 3 at 5 LMH) but the performance is lower compared to operating Viresolve® Pro Device filters at flux ≥ 60 LMH (LRV ≥ 6).

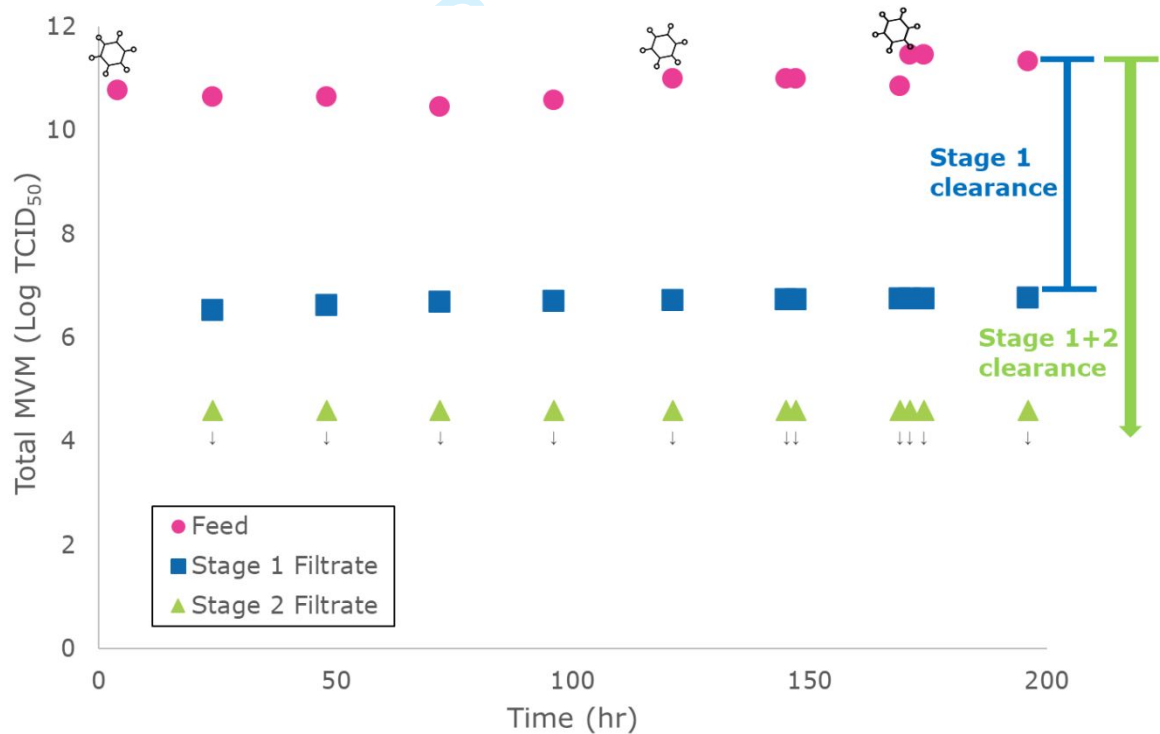


Figure 4. Cumulative MVM load (log TCID₅₀) as a function of filtration time. Virus symbols indicate time of the virus spikes. Downward arrows indicate no virus detected, and the plotted values represent the 95% confidence of equal or less than values.

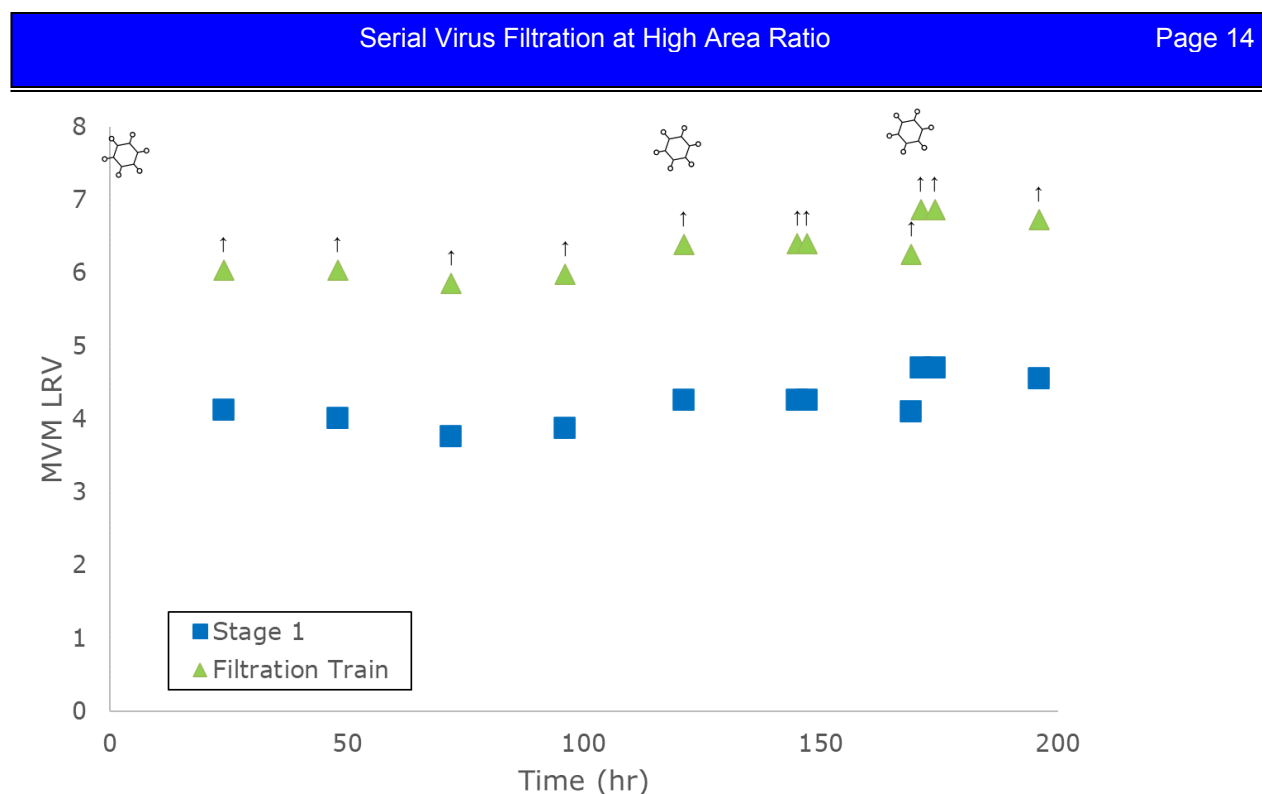


Figure 5. MVM clearance performance in the filtration train over the operation duration. Virus symbols indicate time of the virus spikes. Upward arrows indicate virus clearance performance is greater than or equal to values at a 95% confidence.

There was no MVM detected in the second stage filtrate over the filtration operation including immediately after process interrupts. Based on the sample fractions of the filtration train pools, the total virus in the second stage filtrate at the end of the run was ≤ 4.6 log TCID₅₀, based on a 95% probability. The total MVM that challenged the filtration train was 11.3 log TCID₅₀, resulting in a total LRV across the filtration train ≥ 6.7 at the end of the run. Based on the LRV vs flux data in Figure 3, the expected total LRV of the filtration train was >9 , but the measured LRV was limited by virus challenge level and filtrate sampling due to operation scale. This study highlights that a serial filtration train using Viresolve® Pro filters at a high area ratio can demonstrate robust MVM clearance performance under continuous operation flux conditions including process interruptions.

4.3 Process simulation

The experimental results of this study along with the operating conditions, filter properties, and the commercially available Viresolve® Pro Device filter sizes listed in Table 2 were used as the basis for estimating virus filtration performance within a commercial scale continuous biomanufacturing process. The unfouled membrane $R_0 \cdot A$ as measured in this study was used in the simulation but since the study feed stream was formulated to be higher fouling than streams that are typically virus filtered, lower fouling parameters were used in the simulation. While filters within each stage can be manifolded to create a very large number of area ratio combinations, for purposes of simplicity this simulation was limited to a single filter in each stage, selectable from the commercially available filter sizes listed in Table 2.

Table 2. Operating conditions, filtration properties, and EFA of commercially available Viresolve® Pro filter sizes. The K_i values translate to V90 values of 1000 and 100,000 L/m² for the first and second stages respectively, using the intermediate model.

Operating Conditions

Flow Rate (L/h)	10
Max ΔP (kPa)	138

Filtration Properties

Property	Stage 1	Stage 2
R_0/A (kPa/LMH)	0.42	0.42
K_i (m ⁻¹)	2.3	0.023

Filter Sizes

Name	EFA (m ²)
Viresolve® Pro Modus 1.1 Device	0.017
Viresolve® Pro Modus 1.2 Device	0.07
Viresolve® Pro Modus 1.3 Device	0.22
Viresolve® Pro Magnus 2.1 Device	0.51
Viresolve® Pro Magnus 2.2 Device	1.53

Virus retention performance for each stage was based on fit to the data shown in Figure 3, per the following empirical equation:

$$LRV_0 = 2.69 * \log(J) + 1.62 \text{ for } J \leq 60 \text{ LMH and } LRV_0 = 6.4 \text{ for } J \geq 60 \text{ LMH} \tag{9}$$

Where:

J = flux (LMH)

Virus retention across both stages can be calculated by substituting Eq. 9 into Eq. 8 and using the parameters listed in Table 2. The pressure drop across each stage can be calculated by applying Eq. 4 to each stage, with the total pressure drop across the filtration train equal to the sum of the pressure drops across each stage.

Simulation results for several single stage and serial virus filtration scenarios are summarized in Table 3. For single stage operation, low flux operation allows for maximum productivity (in terms of volume filtered per filtration area) and long changeout times (based on ΔP and permeability decay constraints), but with a virus retention performance penalty. For example, running the process with a single Magnus 2.2 device (case 1 in Table 3) allows for a filter changeout time of 153 hours and a filter productivity of 1000 L/m², but the projected virus clearance performance is unacceptably low (LRV ~ 3). A much smaller filter operated at a much higher flux (case 4) results in an acceptable LRV of 6, but the productivity is much lower and the changeout time is only 3 hours, rendering essentially a batch filtration process.

The retention vs flux and changeout time tradeoff can be significantly shifted using a serial filtration configuration. By connecting a Viresolve® Pro Modus 1.2 Device to the Viresolve® Pro Magnus 2.2 Device of case 1 (which becomes case 6), LRV is projected to increase from about 3 to about 9, without any decrease in the filter changeout time of 153 hours and only a small overall productivity decrease (-5%). Because there is an excess of required LRV, this configuration can allow for an extension of filter life without risking a virus retention shortfall. For example, increasing the allowable flux decay from 90% to 95% in case 5 (which becomes case 9) increases the filtration train productivity and filter changeout time by about 30% with only a small decrease in projected LRV which is still well above typical assay limits. Compared to single stage operation at the same stage 1 flux (case 1), case 9 is projected to have about 30% longer changeout time, 14% higher productivity, and nearly 6 LRV higher. Compared to single stage operation where the LRV ≥ 6 (case 4), case 9 is projected to have about 75X longer changeout time, 3X higher productivity, and > 2 LRV higher.

Table 3. Calculated virus filtration performance. Filter life (changeout time) based on maximum allowable permeability decay, or a maximum pressure drop across the filtration train of 138 kPa, whichever is reached first.

Case No.	Filter type		Max Perm Decay (%)	Area Ratio	Flux (LMH)		Final ΔP @ changeout (kPa)			1-Ro/R @ changeout		V/A (L/m ²)	LRV		
	Stg 1	Stg 2			Stg 1	Stg 2	Stg 1	Stg 2	Total	Stg 1	Stg 2		Stg 1	Stg 2	Total
1	Magnus 2.2	None	90	N/A	7	N/A	27	N/A	27	0.90	N/A	1000	2.8	N/A	2.8
2	Magnus 2.1	None	90	N/A	20	N/A	80	N/A	80	0.90	N/A	1000	4.1	N/A	4.1
3	Modus 1.3	None	90	N/A	45	N/A	138	N/A	138	0.87	N/A	874	5.2	N/A	5.2
4	Modus 1.2	None	90	N/A	143	N/A	138	N/A	138	0.58	N/A	377	6.0	N/A	6.0
5	Magnus 2.2	Modus 1.3	90	7.0	7	45	27	22	48	0.90	0.15	874	2.8	6.0	8.8
6	Magnus 2.2	Modus 1.2	90	21.9	7	143	27	96	122	0.90	0.40	956	2.8	6.4	9.2
7	Magnus 2.1	Modus 1.3	90	2.3	20	45	80	19	99	0.90	0.05	699	4.1	6.1	10.2
8	Magnus 2.1	Modus 1.2	90	7.3	20	143	70	68	138	0.89	0.15	831	4.2	6.4	10.6
9	Magnus 2.2	Modus 1.3	95	7.0	7	45	53	23	76	0.95	0.19	1137	2.5	6.0	8.5
10	Magnus 2.2	Modus 1.2	95	21.9	7	143	36	102	138	0.93	0.43	1079	2.7	6.4	9.1

The permeability decay of the second stage filter in case 9 is only about 19%, well below its maximum useful life. Therefore, this filter could continue to be used when the first stage filter is changed out, further increasing total filter productivity. For case 9, it is calculated that the second stage filter would need to be changed out after every 3rd changeout of the first stage filter (597 hours of total operating time) to not exceed the maximum allowable filtration train ΔP of 138 kPa. As a result, total filtration train productivity increases by 9% compared to changing out the second stage filter with every changeout of the first stage filter, while maintaining an LRV > 8. This methodology provides substantial virus safety (higher LRV) and economic advantages (higher productivity) over single stage operations at low flux.

5. Conclusions

While continuous processes offer numerous advantages over batch operations, integrating traditional batch operations seamlessly remains challenging. Virus filtration in continuous processes often operates at low flux, which can compromise virus retention performance. In this study, MVM retention performance of a Viresolve[®] Pro Device at 5 LMH, representative of the low end of the flux range in continuous operation, was about 2 LRV lower than retention performance at 60 LMH or higher, where virus filters in batch operation are typically operated. This parvovirus retention performance penalty has inhibited the adoption of truly continuous virus filtration. Although virus filtration can be employed within a continuous operation as a sub-batch process, the requirement for surge tanks upstream and downstream of the virus filter and relatively frequent changeout of the virus filter diminish the advantages of continuous operation.

To overcome the retention performance loss associated with low flux typical of continuous virus filtration, a serial filtration approach at high first stage to second stage area ratio is introduced. At high area ratio, the second stage operates at a higher flux (proportional to the area ratio) than the first stage and therefore does not suffer a parvovirus retention performance reduction as long as the second stage flux is sufficiently high (>50 LMH). Since the LRV of the filtration train is the sum of the LRV of each stage in the train, the total LRV of such a serial filtration configuration is significantly higher than a single stage operation. While adding a second stage adds cost, the amount of added filtration area at high area ratio is relatively small. In this study, virus filters were operated continuously (except for three planned process interruptions) for 200 hours in a serial configuration at a first to second stage area ratio of 13:1 at a first stage flux of 5 LMH. While Viresolve[®] Pro Device MVM clearance performance was about 4 LRV across the first stage filter, there was no virus detected in the second stage filtrate, demonstrating robust MVM clearance performance using the filtration train (LRV ≥ 6.7). The second stage filter was the dominant flow resistance at the start of the run but, as it was protected from foulants by the first stage filter, suffered minimal fouling and the life of the filter train (determined by a maximum allowable pressure drop across the filtration train) was controlled by the first stage filter.

Simulations performed for a hypothetical continuous bioprocessing scenario indicated that further performance advantages of serial virus filtration at high area ratio are possible. In single stage operation, filter life can be limited by the imposition of a maximum permeability decay constraint, to protect against LRV loss when the filter reaches a very high degree of fouling. However, using a serial virus filtration train, which demonstrates highly robust MVM clearance performance, the life of the virus filters can be extended without risking a virus retention shortfall. Allowing 95% permeability decay of the first stage filter in a serial

configuration was projected to have about 30% longer changeout time, 14% higher productivity, and virus retention nearly 6 LRV higher than single stage operation where the allowable permeability decay was 90%. Since the second stage filter in a dual stage configuration will generally experience minimal fouling, the second stage filter can potentially be used for multiple changeout cycles of the first stage filter, further increasing overall virus filter productivity. Continuous operations which adopt this methodology over a sub-batch virus filtration process will trade frequent changeouts and surge tanks for adding one, relatively small, virus filter which enables superior virus retention and virus filtration productivity.

Author Contributions

Sal Giglia: Conceptualization (lead); formal analysis (equal); methodology (equal); supervision (lead); visualization (equal) writing - original draft (lead); writing - review and editing (equal). **Ben Cacace:** Conceptualization (supporting); formal analysis (equal); methodology (equal); supervision (supporting); writing - original draft (equal); writing - review and editing (equal). **Jacob McCoskey:** Formal analysis (equal), investigation (equal); methodology (equal); resources (equal); writing - original draft (equal); writing - review & editing (equal). **Matthew Tierson:** Formal analysis (equal), investigation (equal); methodology (equal); resources (equal); writing - original draft (equal); writing - review & editing (equal). **Christina Carbrelo:** Formal analysis (supporting), writing - review and editing (supporting). **Corinne Miller:** Methodology (supporting); writing - review and editing (supporting). **Joseph Hersey:** Conceptualization (supporting); methodology (equal); supervision (supporting); writing - review and editing (equal).

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